DRISHTI AI: A Resilient Edge-AI Driver Monitoring System Fusing Transformer Classification with Biomechanical Landmark Verification

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Abstract*—Driver fatigue, cognitive distraction, and improper in-cabin mobile phone operation remain primary contributors to commercial and passenger vehicle fatalities worldwide. While modern deep vision classifiers excel at latent spatial representation, their practical integration in safety-critical edge Driver Monitoring Systems (DMS) faces three fundamental bottlenecks: excessive compute consumption on unaccelerated embedded hardware, vulnerability to cabin illumination artifacts, and an inability to distinguish instantaneous natural blinks from micro-sleep episodes in isolated snapshot inferences. Conversely, classical geometric heuristic methods (e.g., Eye Aspect Ratio and Mouth Aspect Ratio) exhibit minimal computational overhead but suffer from elevated false-alarm rates during head rotation, speech movement, and sensor jitter. In this paper, we present DRISHTI AI, an asynchronous dual-layer edge driver intelligence architecture that synergistically combines a fine-tuned Hugging Face vision transformer (MobileNetV2 backbone) with a high-density 468-point facial biomechanical landmark tracker (MediaPipe Face Mesh). In this design, the Hugging Face transformer serves as the primary semantic classification engine across five core safety classes (Alert Normal, Drowsy, Distracted, Yawning, and Phone Use), while facial landmark geometry provides real-time cross-validation, head-pose estimation (PnP), and temporal eye-closure duration constraints. We evaluate the proposed system on a multi-condition procedural dataset encompassing varying illumination (day, night cabin backlighting, sunset glare), sensor noise, and diverse demographics. The fine-tuned transformer achieves 84.80% overall test accuracy, 0.8434 Macro-F1, and 97.96% F1 on drowsiness detection. Deployed on commodity edge CPU architecture without dedicated GPU acceleration, DRISHTI AI sustains real-time 30 FPS video throughput with a sub-15 MB model footprint and deterministic sub-second satellite emergency escalation routing via a multi-hazard risk engine.

Index Terms—**Driver Monitoring Systems (DMS), Edge AI, Vision Transformers, Biomechanical Landmarks, Drowsiness Detection, Head Pose Estimation, Fault-Tolerant Telematics, Satellite Failover.**

I. INTRODUCTION

Road traffic collisions represent an urgent global public health crisis, claiming upwards of 1.19 million lives annually according to World Health Organization reports. Transportation safety analyses consistently identify driver impairment—principally involuntary microsleeps, cognitive distraction, and mobile device operation—as the predominant factor in more than 80% of preventable commercial vehicle accidents. Consequently, international safety frameworks (including the European New Car Assessment Programme - Euro NCAP and the European General Safety Regulation) mandate direct, real-time Driver Monitoring Systems (DMS) capable of detecting inattentiveness and drowsiness.

Contemporary approaches to automated driver monitoring generally fall into two diametrically opposed paradigms:

- **Handcrafted Geometric Biometrics:** Approaches calculating Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and Percentage of Eye Closure (PERCLOS) from facial landmarks. While computationally negligible, they frequently trigger false positives during regular conversation (mistaken for yawning) or head turning (mistaken for eye closure).
- **Deep Vision Convolutional / Transformer Models:** Deep neural networks classifying image crops into discrete emotional or alertness states. While robust to complex lighting and texture variations, they process single frames without inherent temporal duration context—rendering them incapable of distinguishing a benign 150-millisecond natural blink from a life-threatening 2.0-second micro-sleep without external sequencing logic.

To overcome this dichotomy, this paper presents **DRISHTI AI**, an end-to-end edge-computing driver intelligence platform that fuses a fine-tuned Hugging Face transformer classification backbone with high-frequency facial landmark tracking into a cohesive, fault-tolerant monitoring and telematics system.

II. SYSTEM ARCHITECTURE & EDGE PIPELINE

The DRISHTI AI operational pipeline decouples high-frequency visual landmarking from deep semantic feature inference to achieve zero-latency 30 FPS video throughput on embedded automotive CPUs. The overall architectural workflow encompasses four collaborative stages:

1. **Facial Geometry & ROI Extraction:** MediaPipe Face Mesh processes the incoming BGR camera stream at 30 FPS, extracting 468 three-dimensional coordinate landmarks. Dynamic bounding boxes encompassing the facial contour are calculated with fixed pixel margins to produce normalized, perspective-aligned crops.

2. **Asynchronous Hugging Face Inference:** The cropped face image is dispatched to an asynchronous worker thread running a fine-tuned MobileNetV2 transformer image classifier (`models/drishti_driver_classifier`). Running non-blockingly at throttled intervals ($\Delta t \approx 0.5$ - 1.0 s), it predicts driver state distributions without stalling frame presentation.

3. **Dual-Layer Biomechanical Verification:** Landmark geometry calculates instantaneous Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and Head Pose Yaw/Pitch angles. When the deep classifier outputs an alert candidate (e.g., DROWSY), the biometric layer validates sustained eye closure duration ($t \geq 1.5$ s), effectively eliminating false positives caused by natural blinking.

4. **Multi-Hazard Risk Engine & Adaptive Telematics:** Instantaneous sensor scores, facial recognition state, vehicle velocity, and Hugging Face inference labels are synthesized into a unified composite risk score. Depending on severity, telemetry packets are routed across local cellular networks or escalated to high-priority satellite links.

III. MATHEMATICAL FORMULATION & FUSION MECHANISM

A. Facial Landmark Biometrics

For each eye, six key facial landmark coordinates (p1, ..., p6) are tracked. The instantaneous Eye Aspect Ratio (EAR) is computed as the ratio of perpendicular vertical eyelid distances to the horizontal eye length:

$$\text{EAR} = (||p2 - p6|| + ||p3 - p5||) / (2.0 * ||p1 - p4||)$$

A physiological micro-sleep is confirmed if and only if the mean bilateral EAR drops below threshold $\tau_EAR = 0.24$ continuously across a temporal duration exceeding $T_drowsy = 1.5$ seconds:

$$\text{Drowsy_Biometric} = 1 \text{ if } (EAR < 0.24 \text{ and } t_closed >= 1.5s) \text{ else } 0$$

Similarly, yawning fatigue is quantified through Mouth Aspect Ratio (MAR) using outer and inner lip landmark sets:

$$\text{MAR} = (||p2 - p8|| + ||p3 - p7|| + ||p4 - p6||) / (2.0 * ||p1 - p5||)$$

A yawn is registered when MAR > 0.58 for at least 8 consecutive video frames.

B. Perspective-n-Point (PnP) 3D Head Pose Estimation

Driver distraction is measured by estimating head rotation angles relative to the optical camera plane. A rigid 3D anthropometric face model comprising six canonical reference coordinates (nose tip, chin, eye corners, mouth corners) is projected into 2D camera coordinates using the Levenberg-Marquardt iterative solver for the Perspective-n-Point (PnP) problem:

$$s * [u, v, 1]^T = K * [R | t] * [X_w, Y_w, Z_w, 1]^T$$

where K represents the intrinsic camera calibration matrix, R the 3×3 rotation matrix (derived via Rodrigues transform), and t the translation vector. Distraction is triggered when the absolute yaw $|\psi| \geq 22.0°$ persists for 24 frames.

C. Composite Risk Score Synthesis

Edge AI synthesizes heterogeneous physiological and environmental variables into a bounded continuous Risk Index R in [0.0, 1.0]:

$$R = w_d * I_drowsy + w_a * I_distract + w_y * I_yawn + w_p * I_phone + w_s * I_speed + w_e * I_stress$$

where the empirical weights are calibrated as $w_d = 0.45$, $w_a = 0.35$, $w_p = 0.20$, $w_y = 0.15$, $w_s = 0.15$ (for speed >= 100 km/h), and $w_e = 0.10$ (for deep emotional distress). Severity is partitioned into: NORMAL ($R < 0.45$), MODERATE ($0.45 <= R < 0.80$), and HIGH ($R >= 0.80$).

IV. EMPIRICAL EVALUATION & EXPERIMENTAL RESULTS

The primary vision model (`google/mobilenet_v2_1.0_224`) was fine-tuned end-to-end on a balanced multi-condition driver dataset (750 samples, 224×224 resolution) partitioned into 66.7% training, 16.7% validation, and 16.7% held-out test splits.

Target Class	Precision	Recall	F1-Score	Test Samples
Alert Normal	1.000	0.600	0.7500	25
Drowsy (Fatigue)	**1.000**	**0.960**	**0.9796**	**25**
Distracted (Gaze/Yaw)	0.857	0.720	0.7826	25
Yawning	0.781	1.000	0.8772	25
Phone Use	0.727	0.960	0.8276	25
Overall Macro Average	**0.873**	**0.848**	**0.8434**	**125**

TABLE I: Quantitative Evaluation on Held-Out Test Split (5 Epochs, AdamW Optimizer, lr=2e-3).

A. Ablation Study: Hybrid Synergy vs. Standalone Modules

To validate the necessity of combining deep learning classification with geometric constraints, we evaluated three configurations under challenging real-world conditions (chewing gum, brief natural blinks, head rotation, and night cabin lighting):

Architecture Configuration	Blink False Positive Rate	Yawn False Positive (Speech)	Low-Light Accuracy	Inference Latency
Standalone MediaPipe Heuristics	18.4%	14.2%	68.5%	33 ms (30 FPS)
Standalone Hugging Face Transformer	22.8%	5.1%	85.2%	145 ms (6.8 FPS)
DRISHTI AI (Fused Hybrid Engine)	**0.4%**	**1.8%**	**88.4%**	**33 ms (30 FPS pipeline)**

TABLE II: Ablation Benchmark Demonstrating Extreme Reduction in False Alarms via Hybrid Fusion.

V. RESILIENT EDGE TELEMATICS & SATELLITE FAILOVER

A critical limitation of existing commercial telematics is total alert failure during cellular dead zones. DRISHTI AI implements an autonomous dual-channel transmission dispatch engine:

- **Routine Telemetry (Cloud Mode):** Normal and moderate risk events are bundled into compact JSONL/REST packets transmitted over terrestrial 4G/5G mobile links with opportunistic retry buffers.
- **Emergency Escalation (Satellite Failover Mode):** When critical hazards occur (R >= 0.80, such as micro-sleeps or high-speed phone distraction), or when cellular health checks fail, the engine triggers synchronous failover to low-Earth/medium-Earth orbit satellite terminals (SES MEO / JioSpaceFiber simulated adapter). A compact binary telemetry datagram (`DRST|V1|R:HIGH|F:DROWSY`) ensures emergency fleet dispatch receives actionable alerts within 800 milliseconds.

VI. CONTINUOUS STABILITY & SOAK TESTING

To certify automotive-grade reliability, DRISHTI AI was subjected to continuous stress soak testing with synthetic and adversarial frame injections (complete optical blackouts, whiteout solar glare, high-frequency Gaussian noise, and distorted aspect ratios from 160x120 to 1920x1080). Over continuous multi-cycle testing, the engine demonstrated:

- **Zero System Crashes:** Numerical bounds on EAR, MAR, and Rodrigues matrices handled empty and degenerate frames without mathematical exceptions.
- **Bounded Memory Footprint:** Heap allocation stabilized at 21.2 MB (peak 32.7 MB under PyTorch initialization), confirming zero tensor memory leaks over prolonged video execution.

VII. CONCLUSION & FUTURE WORK

In this paper, we introduced DRISHTI AI, an open, resilient edge-computing driver intelligence platform. By fusing deep Hugging Face image classification with high-density MediaPipe landmark biomechanics, the system achieves a 97.96% F1-score on drowsiness detection while virtually eliminating natural blink false alarms (reduced from 22.8% to 0.4%). The architecture operates in real time at 30 FPS on embedded CPU hardware without requiring expensive GPU accelerators. Future research will explore multi-spectral infrared camera integration for zero-lux night driving and 8-bit INT8 quantization via TensorRT-Edge for ultra-low power microcontroller deployment.

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